

7. [Request] Request Body Multipart Form Data

Multipart Form Data কী?

Multipart Form Data হলো এমন এক ধরনের request body যেটা দিয়ে ফাইল + টেক্সট ডাটা একসাথে API-তে পাঠানো যায়।

- Image upload
- Profile picture
- PDF / document upload
- Form + File একসাথে পাঠাতে
- (Three Dots) মানে কী?
- মানে Required field (আবশ্যক)
- মানে এই ফিল্ড না দিলে request fail করবে

```
pip install python-multipart
```

Simple Form Data Example

```
from fastapi import FastAPI, Form
app = FastAPI()
@app.post("/login")
def login( email: str = Form(...), password: str = Form(...)):
    return {
        "email": email,
        "password":password
    }
```

Catch Form Data Direct Using Request

```
from fastapi import FastAPI, Request

app = FastAPI()

@app.post("/form")
async def get_form_data(request: Request):
    form_data = await request.form()
    return dict(form_data)
```

```
from fastapi import FastAPI, UploadFile, File
import uvicorn
app = FastAPI()

@app.post("/hello")
async def upload_file(file: UploadFile = File(...)):
    data = await file.read()    # read in memory
```

```

    return {
        "filename": file.filename,
        "content_type": file.content_type,
        "size_bytes": len(data),
        "size_kb": round(len(data) / 1024, 2),
        "headers": dict(file.headers)
    }

if __name__ == "__main__":
    uvicorn.run("main:app", host="127.0.0.1", port=8000, reload=True)

```

File Upload Example

```

from fastapi import FastAPI, UploadFile, File
import shutil
import os

app = FastAPI()

UPLOAD_DIR = "uploads"
os.makedirs(UPLOAD_DIR, exist_ok=True)

@app.post("/upload")
async def upload_file(file: UploadFile = File(...)):
    file_path = f"{UPLOAD_DIR}/{file.filename}"

    with open(file_path, "wb") as f:
        shutil.copyfileobj(file.file, f)

    return {
        "filename": file.filename,
        "content_type": file.content_type,
        "message": "File uploaded successfully"
    }

```